

1 **Dissecting the cardiometabolic and inflammatory**
2 **consequences of genetically proxied NLRP3 activity for**
3 **coronary artery disease risk**

4 Nick Hirschmüller¹,, Stephen Burgess^{1,2}, Dirk S. Paul³, Adam S.
5 Butterworth^{1,4}

6 ¹ British Heart Foundation Cardiovascular Epidemiology Unit, Department of Public Health and Primary Care,
7 University of Cambridge, Cambridge, UK

8 ² MRC Biostatistics Unit, University of Cambridge, Cambridge, UK

9 ³ Centre for Genomics Research, Discovery Sciences, BioPharmaceuticals R&D, AstraZeneca, Cambridge, UK

10 ⁴ BHF Centre of Research Excellence, University of Cambridge, Cambridge, UK
11

1. Abstract

Background and Aims

Preclinical studies implicate NLRP3 inflammasome activation, which drives release of interleukin-1 β and interleukin-18, in vascular inflammation and atherosclerosis, and selective NLRP3 inhibitors are now in clinical development. However, whether lowering NLRP3 activity reduces coronary artery disease (CAD) risk in humans is unknown. We used human genetics to estimate this effect and the pathways mediating it.

Methods

A genetic NLRP3 activity score was derived by combining the effects of eight variants at the *NLRP3* locus on *NLRP3* expression and three inflammatory biomarkers: C-reactive protein, glycoprotein acetylation (GlycA), and neutrophil count. The score was validated against circulating interleukin-1 β and interleukin-18. Drug-target Mendelian randomization was used to estimate the effect of lower NLRP3 activity on CAD risk, cardiometabolic traits, and other diseases for which NLRP3 inhibitors are in development. Mediation by cardiometabolic risk factors was quantified using the product-of-coefficients method.

Results

A lower genetically proxied NLRP3 activity score was associated with lower circulating interleukin-1 β and interleukin-18, but with higher CAD risk (OR = 1.21, 95% CI 1.02–1.45, P = 0.032). It was also associated with higher systolic blood pressure, apolipoprotein B levels, and type 2 diabetes risk, which together mediated an estimated 87% of the association with CAD. The residual direct association with CAD was attenuated (OR = 1.03, 95% CI 0.85–1.24, P = 0.79). In contrast, a lower score was nominally associated with lower risk of gout, pericarditis, and Parkinson's disease.

Conclusions

Genetically proxied lower NLRP3 activity was associated with higher CAD risk, largely mediated by adverse cardiometabolic effects, with no evidence of a residual protective effect. These findings reflect lifelong genetic variation and should not be interpreted as equivalent to shorter-term pharmacological inhibition.

2. Introduction

Despite optimal management of established cardiovascular risk factors, including low-density lipoprotein cholesterol (LDL-C) and blood pressure, patients with established coronary artery disease (CAD) remain at considerable risk of recurrent cardiovascular events. Persistent vascular inflammation, reflected by elevated high-sensitivity C-reactive protein levels, is recognised as a major contributor to this residual cardiovascular risk¹. The CANTOS trial provided clinical proof-of-concept that selective inhibition of interleukin-1 β (IL-1 β), a pro-inflammatory cytokine, reduces recurrent cardiovascular events independently of lipid lowering². Trials of the broad anti-inflammatory drug colchicine have yielded mixed results, with benefit observed in LoDoCo2³ among patients with chronic coronary disease, but not in CLEAR-SYNERGY⁴ among patients with acute myocardial infarction. Recently, the phase 3 ZEUS trial found no reduction in cardiovascular events with interleukin-6 (IL-6) inhibition⁵. Together, these findings suggest that cardiovascular benefit depends on both the inflammatory pathway targeted and the clinical context, rather than on inflammation lowering per se.

One inflammatory pathway that has gained considerable attention is centred on NLRP3. NLRP3 is the sensor protein of the NLRP3 inflammasome, a multiprotein complex comprising NLRP3, the adaptor ASC, and caspase-1, whose activation promotes the maturation and release of IL-1 β and interleukin-18 (IL-18)⁶. These cytokines promote downstream IL-6 signalling, which drives the acute-phase response characterised by elevated inflammatory markers such as C-reactive protein (CRP) and GlycA, a composite measure of glycosylated acute-phase proteins^{7,8}. Concurrently, NLRP3 inflammasome activation increases the production of circulating myeloid cells, predominantly neutrophils⁹.

In preclinical models of atherosclerosis, pharmacological NLRP3 inhibition has been reported to reduce vascular inflammation and plaque burden¹⁰. Together with the evidence linking NLRP3 inflammasome-related cytokines to vascular inflammation, these findings have motivated the development of small-molecule inhibitors that act directly on the NLRP3 sensor, several of which have entered clinical testing¹¹. However, no phase 3 trial has yet established whether selective NLRP3 inhibition reduces cardiovascular events in humans.

In the absence of large randomised clinical trials, drug-target Mendelian randomization (MR) can provide genetic evidence on whether a target causally influences disease risk, using naturally occurring variation near drug-target genes to proxy differences in target activity. This approach has precedent in cardiovascular drug development. For example, genetic variation in *PCSK9* supported LDL-C lowering through PCSK9 inhibition as a strategy expected to reduce cardiovascular risk¹². Conversely, genetic proxies of lower lipoprotein-associated phospholipase A₂ (Lp-PLA₂) activity were not associated with lower coronary heart disease risk, consistent with the lack of benefit observed for pharmacological Lp-PLA₂ inhibition¹³.

In this study, we applied drug-target MR to test whether genetically proxied NLRP3 activity is associated with CAD risk, examined the pathways underlying this association, and explored additional diseases for which selective NLRP3 inhibitors have been or are being evaluated in clinical trials.

3. Methods

3.1. Study design

We used drug-target MR to test whether genetically proxied NLRP3 activity, represented by an NLRP3 activity score, was associated with CAD risk (Figure 1). We constructed the score from eight *cis*-acting variants at the *NLRP3* locus, using their associations with *NLRP3* expression and circulating inflammatory biomarkers downstream of NLRP3 inflammasome activation. We validated this score against canonical NLRP3 inflammasome effector cytokines IL-1 β and IL-18 before estimating its association with CAD. We further assessed mediation by cardiometabolic risk factors and explored additional diseases for which selective NLRP3 inhibitors have been or are being evaluated in clinical trials.

3.2. Data sources

Across analyses, we prioritised the largest publicly available GWAS summary statistics from European-ancestry participants to maximise statistical power while maintaining ancestry consistency with the instrument-construction datasets.

3.2.1. Datasets for instrument construction and validation

Instrument construction used GWAS summary statistics for circulating CRP levels, GlycA concentrations, neutrophil count, and whole-blood *NLRP3* expression. Summary statistics for CRP levels were obtained from a UK Biobank and CHARGE meta-analysis including 575,531 European-ancestry individuals, with CRP measured using immunoturbidimetric assays¹⁴. Summary statistics for GlycA concentrations were obtained from 434,646 European-ancestry UK Biobank participants, with GlycA quantified using the Nightingale Health nuclear magnetic resonance metabolomics platform¹⁵. Neutrophil count summary statistics were obtained from the European-ancestry subset of a multi-ancestry blood-cell trait meta-analysis including 519,288 individuals, in which counts were measured by automated haematology analysers in contributing cohorts¹⁶. Whole-blood *NLRP3* expression quantitative trait locus data were obtained from 4,732 European-ancestry INTERVAL participants, with expression measured by RNA sequencing¹⁷. For instrument validation, we used plasma protein GWAS summary statistics from the full UK Biobank Pharma Proteomics Project cohort, comprising up to 51,637 participants per protein profiled using the Olink Explore platform¹⁸. This cohort is predominantly of European ancestry (45,397 of 52,363 participants). We used these full-cohort summary statistics, which the original study provides to maximise statistical power and variant coverage¹⁸. IL-1 β and IL-18 were available only in this smaller proteomic dataset and were therefore used to validate the score rather than to construct it. Further details are provided in Supplementary Table 1.

3.2.2. Datasets for disease outcomes

The primary disease outcome was CAD, for which we combined four European-ancestry datasets: a CAD GWAS meta-analysis including 181,522 cases and 984,168 controls¹⁹, coronary atherosclerosis in the Million Veteran Program (MVP), including 124,302 cases and 300,039 controls²⁰, coronary atherosclerosis in All of Us, including 19,856 cases and 150,307 controls²¹, and coronary atherosclerosis in FinnGen, including 63,307 cases and 416,171 controls²². SNP-CAD association estimates were combined by fixed-effect inverse-variance weighted meta-analysis, and the resulting meta-analysed summary statistics served as the primary CAD outcome for MR. As a measure of subclinical atherosclerosis, we obtained summary statistics for imaging-defined atherosclerosis from the Swedish CARDioPulmonary bioImage Study (SCAPIS), comprising coronary artery calcification (n=26,000), coronary plaque burden quantified as the segment involvement score on CT angiography (n=24,811), and ultrasound-assessed carotid plaque (n=26,807), which have no sample overlap with the datasets used for instrument construction or with the CAD datasets²³. Further details are provided in Supplementary Table 2. We additionally obtained

European-ancestry summary statistics for diseases for which selective NLRP3 inhibitors have been or are being evaluated in clinical trials, with data sources provided in Supplementary Table 3.

3.2.3. Datasets for cardiometabolic and lifestyle traits

We used publicly available European-ancestry GWAS summary statistics for cardiometabolic and lifestyle traits relevant to CAD risk, including lipid traits, blood pressure, body mass index, type 2 diabetes, smoking, and alcohol consumption. These traits were screened for association with the NLRP3 activity score, and those showing evidence of association were carried forward to mediation analysis. Further details are provided in Supplementary Table 4.

3.3. Genetic instrument construction

To construct a genetic proxy for NLRP3 activity, we used whole-blood *NLRP3* expression together with three inflammatory traits that are elevated downstream of NLRP3 inflammasome activation: circulating CRP and GlycA levels, and blood neutrophil count. For each trait, variants within ± 150 kb of *NLRP3* were LD-clumped at $r^2 < 0.1$ using 11,863 European-ancestry whole-genome sequences from INTERVAL as the linkage disequilibrium (LD) reference panel. We combined the clumped variants across traits and retained variants associated at genome-wide significance ($P < 5 \times 10^{-8}$) with at least two traits, yielding eight variants for instrument construction (Supplementary Figure 1). Trait-specific genetic effect estimates were centred and scaled, and principal component analysis (PCA) was applied to the resulting variant-by-trait matrix. Loadings from the first principal component were used as fixed weights to combine variant effects across traits into a single latent factor. This latent factor was defined as the NLRP3 activity score and served as the exposure in subsequent genetic analyses. The score was scaled to CRP, such that a one-unit decrease corresponds approximately to a one standard deviation decrease in CRP. Standard errors were estimated using the delta method²⁴. Instrument strength was summarised by the F-statistic for the eight variants jointly, accounting for their residual LD. As a methodological positive control, we applied the same procedure at the *IL1RN* locus to construct a genetically proxied interleukin-1 receptor antagonist (IL1Ra) activity score and tested it against plasma IL1Ra concentration and against gout and rheumatoid arthritis, two diseases treated with recombinant IL1Ra (anakinra).

3.4. Colocalisation analyses

To evaluate whether inflammatory biomarker associations at the *NLRP3* locus share a causal variant, we performed multi-trait colocalisation using HyPrColoc²⁵ across whole-blood *NLRP3* expression, circulating CRP levels, GlycA concentrations, and blood neutrophil count, using variants within ± 200 kb of *NLRP3*. Because the *NLRP3* expression association contained two distinct signals, we used GCTA-COJO²⁶ with the INTERVAL European-ancestry whole-genome sequencing panel as the LD reference. We conditioned the *NLRP3* expression summary statistics on the lead variant of the narrow upstream peak (rs6689545) to isolate the association cluster within the *NLRP3* gene. The conditioned *NLRP3* expression summary statistics, together with the unconditioned summary statistics for CRP, GlycA, and neutrophil count, were then used as input for multi-trait colocalisation. HyPrColoc was then run using default variant-specific priors (prior.1 = 1×10^{-4} , prior.c = 0.02). Evidence for colocalisation was summarised using the posterior probability that the traits share a causal variant, with posterior probability > 0.8 interpreted as strong evidence of colocalisation. To test for colocalising signals elsewhere in the region, we repeated HyPrColoc, with the same inputs and priors, in the flanking regions from 200 kb to 1 Mb upstream and downstream of *NLRP3*.

3.5. Mendelian randomization analyses

We used two-sample MR to estimate associations of the NLRP3 activity score with circulating biomarkers, cardiometabolic traits, and disease outcomes. All estimates were oriented to represent lower genetically proxied NLRP3 activity, mirroring pharmacological inhibition. The primary estimator was

random-effects inverse-variance weighted (IVW) MR. Weighted median MR was used as a robustness analysis and the MR-Egger intercept was used to test whether variant effects on the outcome were proportional to their effects on the exposure. MR-Egger was not used as a causal estimator, as its InSIDE assumption is not well suited to variants within a single gene region²⁷. IVW and MR-Egger accounted for residual LD using the INTERVAL reference panel. Heterogeneity was assessed with Cochran's Q. For CAD, we additionally performed leave-one-out analyses, estimated a Wald ratio for the shared colocalising variant (rs12239046) alone, and repeated instrument selection and score construction at relaxed LD clumping thresholds (r^2 0.2–0.6). Estimates are reported as betas (continuous outcomes) or odds ratios (binary outcomes) with 95% confidence intervals. Analyses used the MendelianRandomization²⁸ (version 0.10.0) and MVMR²⁹ (version 0.4.2) R packages.

3.6. Evaluation of the NLRP3 activity score

We first assessed internal consistency by estimating associations of a lower NLRP3 activity score with the downstream inflammatory biomarkers used for score construction: circulating CRP levels, GlycA concentrations, and blood neutrophil count. We then evaluated circulating IL-1 β , IL-18, and IL-6 levels as external biological readouts. IL-1 β and IL-18 were selected because they are canonical cytokine products of NLRP3 inflammasome activation, while IL-6 was included as a downstream cytokine linking interleukin-1 signalling to the acute-phase response^{6–8}.

3.7. Mediation analysis

To investigate whether the association between a lower genetically proxied NLRP3 activity score and higher CAD risk could be explained by established cardiovascular risk factors, we performed mediation analyses for systolic blood pressure (SBP), apolipoprotein B (ApoB), and type 2 diabetes (T2D). These traits were selected because they were associated with the NLRP3 activity score in the cardiometabolic and lifestyle trait screen and are established risk factors for CAD³⁰. ApoB was selected as the lipid mediator because it reflects the concentration of atherogenic ApoB-containing lipoprotein particles and provides a parsimonious measure of lipid-related cardiovascular risk, avoiding joint modelling of highly correlated lipid fractions³¹. We used the product-of-coefficients method to estimate indirect effects³². First, we estimated the effect of a lower genetically proxied NLRP3 activity score on each mediator using two-sample MR. Second, we estimated the mutually adjusted effects of SBP, ApoB, and T2D on CAD risk using multivariable Mendelian randomization (MVMR)²⁹. In this step, SBP, ApoB, and T2D were instrumented using genome-wide significant variants from their respective source GWAS^{15,33,34}, independently of the cis-acting NLRP3 variants used to instrument the activity score. Variants were clumped within each trait and then again across traits ($r^2 < 0.001$, 10 Mb window, INTERVAL reference panel), so that the combined instrument set is approximately independent. Multiplicative random effects were applied to account for heterogeneity across the mediator instruments. The mediator-to-CAD effects and total NLRP3-to-CAD effect used in the mediation analysis were derived from the meta-analysis of the four CAD datasets. The indirect effect through each mediator was calculated as the product of the NLRP3-to-mediator effect and the corresponding mediator-to-CAD effect. The joint indirect effect was calculated as the sum of these mediator-specific indirect effects. Because SBP, ApoB, and T2D are correlated, uncertainty in the joint indirect effect was estimated using the multivariate delta method²⁴. Instrument strength in the multivariable model was assessed using the conditional F-statistic²⁹. Because the mediator genome-wide association studies (GWAS) draw on partly overlapping samples, the covariance between their variant effect estimates was estimated from z-scores at variants that were null for both traits ($P > 0.05$), taking one variant per 100 kb genome-wide. The residual direct effect was defined as the effect of a lower genetically proxied NLRP3 activity score on CAD risk not explained by these three mediators and was estimated by subtracting the joint indirect effect from the total NLRP3-to-CAD effect. All mediation analyses were performed on the log-odds scale.

3.8. *NLRP3* Predicted Loss-of-Function Analysis

To test whether rare *NLRP3* predicted loss-of-function (pLOF) variants were associated with circulating inflammatory readouts, we performed gene-based burden tests for CRP levels, GlycA concentrations, and blood neutrophil count among UK Biobank participants with whole exome sequencing (WES) data and available biomarker measurements. Burden tests were performed using SAIGE-GENE+ (version 1.5.0)³⁵. pLOF variants were defined as frameshift, stop-gained, splice donor, or splice acceptor variants using variant consequences reported in gnomAD (version 4.1)³⁶. Analyses were restricted to individuals of genetically inferred European ancestry³⁷. Analogous pLOF burden analyses were performed for all protein-coding genes within ± 150 kb of *NLRP3*. Phenotype processing is described in the Supplementary Information.

3.9. *NLRP3* Predicted Gain-of-Function Analysis

Pathogenic gain-of-function variants in *NLRP3* were obtained from the INFEVERS registry³⁸ and Cosson *et al.*³⁹. Carriers were identified among UK Biobank participants with WES data and available biomarker measurements. Only one of these gain-of-function variants was observed in UK Biobank, corresponding to GRCh38 chr1:247424765 C>T (p.Ala439Val). All three carriers were heterozygous. Due to the limited number of carriers, CRP levels, GlycA concentrations, and blood neutrophil counts were reported as percentiles relative to non-carriers. Phenotype processing is described in the Supplementary Information.

3.10. Proteome-wide association and over-representation analyses

To characterise broader circulating protein changes associated with lower genetically proxied *NLRP3* activity, we performed two-sample MR using plasma protein GWAS summary statistics from the UK Biobank Pharma Proteomics Project¹⁸. Associations were estimated using IVW MR with LD correction, as described above. Proteins were considered significant using a Bonferroni correction based on the effective number of independent protein tests⁴⁰, yielding a threshold of $P < 2.7 \times 10^{-5}$. Proteins whose circulating levels were significantly lower or higher with lower genetically proxied *NLRP3* activity were tested separately for enrichment of Gene Ontology Biological Process terms. Details of the effective test correction and over-representation analysis are provided in the Supplementary Information.

3.11. Exploratory analyses across diseases under investigation for *NLRP3* inhibition

Disease outcomes for exploratory MR analyses were selected based on diseases for which selective *NLRP3* inhibitors have been, or are being, evaluated in registered clinical trials, as identified from a structured search of four clinical trial registries conducted on the 28 of April 2026 (see Supplementary Information for full details). Identified diseases were included where publicly available GWAS summary statistics of sufficient sample size were available. The included exploratory outcomes and corresponding GWAS sources are provided in Supplementary Table 3. As these analyses were exploratory, nominal P values are reported without adjustment for multiple testing.

4. Results

4.1. Genetic associations with inflammatory biomarkers at the *NLRP3* locus

Using publicly available GWAS summary statistics (Supplementary Table 1), we identified strong association signals at the *NLRP3* locus with whole-blood *NLRP3* expression (rs58546652, $P = 1.04 \times 10^{-58}$, $n = 4,732$), circulating CRP levels (rs12239046, $P = 1.34 \times 10^{-87}$, $n = 575,531$), GlycA concentrations (rs56188865, $P = 4.7 \times 10^{-28}$, $n = 434,646$), and blood neutrophil count (rs56188865, $P = 1.93 \times 10^{-43}$, $n = 519,288$) (Figure 2A). Multi-trait colocalisation across *NLRP3* expression and the three circulating inflammatory biomarkers showed strong evidence of a shared causal variant between the four traits (posterior probability = 0.97). No additional colocalising signals were observed in the flanking regions between 200 kb and 1 Mb on either side of *NLRP3*, indicating that the inflammatory biomarker associations shared across traits in this region are centred on *NLRP3* (Supplementary Figure 2). Together, these findings support coherent *NLRP3*-linked inflammatory signal at the locus and identify variants suitable for genetic instrument construction.

4.2. Genetic instrument selection and validation

Because no single circulating biomarker specifically measures *NLRP3* activity, we constructed a composite genetic exposure combining whole-blood *NLRP3* expression with downstream circulating inflammatory readouts. We identified genetic signals associated with at least two of the four traits ($P < 5 \times 10^{-8}$). From these signals, we selected eight approximately independent variants ($r^2 < 0.1$, Supplementary Figure 1). We then applied PCA to their effect estimates across the four traits. The first principal component explained 96.8% of the variation across the traits' genetic associations. We used the weights derived from this component to combine these associations into the *NLRP3* activity score (Figure 2B, Supplementary Table 5), which served as the exposure in subsequent MR analyses. The eight variants explained 8.3% of the variance in *NLRP3* expression, 0.11% in CRP, 0.046% in GlycA, and 0.064% in neutrophil count, and the joint F-statistic for the score was 92.0.

To evaluate the *NLRP3* activity score, we used two-sample MR to estimate its associations with downstream biomarkers. We first confirmed that a lower genetically proxied *NLRP3* activity score was associated with lower CRP, GlycA, and neutrophil count, consistent with the score's construction (Figure 2C). We then tested associations with *NLRP3*-related cytokines that were not used in score construction. A lower genetically proxied *NLRP3* activity score was associated with lower circulating IL-1 β ($\beta = -0.25$, 95% CI -0.49 to -0.004, $P = 0.046$), IL-18 ($\beta = -0.52$, 95% CI -0.79 to -0.25, $P = 0.00016$) and IL-6 ($\beta = -0.72$, 95% CI -0.95 to -0.48, $P = 2.8 \times 10^{-9}$). Effect estimates were directionally consistent across IVW and weighted median estimators (Supplementary Table 6). Together, these findings support the *NLRP3* activity score as capturing the expected downstream inflammatory consequences of lower *NLRP3* activity.

4.3. Association of the *NLRP3* activity score with coronary artery disease risk

We performed two-sample MR to estimate the association between the genetically proxied *NLRP3* activity score and CAD risk using a fixed-effect meta-analysis of four large European-ancestry GWAS datasets, comprising 388,987 cases and 1,850,685 controls. Despite being associated with lower inflammatory biomarkers, a lower *NLRP3* activity score was associated with higher CAD risk (OR = 1.21, 95% CI 1.02 to 1.45, $P = 0.032$) (Figure 3A). Estimates were directionally concordant across all four contributing datasets and were preserved under the weighted median estimator (OR = 1.23, 95% CI 1.06 to 1.42, $P = 0.005$). The MR-Egger intercept differed from zero (intercept = 0.009, 95% CI 0.002 to 0.016, $P = 0.018$), indicating that the variants' effects on CAD are not fully proportional to their effect on the *NLRP3* activity score (Supplementary Table 7).

We next asked whether this association was also observed for imaging-defined atherosclerosis in an independent population-based imaging cohort. A lower genetically proxied *NLRP3* activity score was

associated with higher coronary artery calcification ($\beta = 0.31$, 95% CI 0.02 to 0.60, $P = 0.034$) and higher coronary plaque burden, measured as the segment involvement score on CT angiography (proportional odds ratio = 2.55, 95% CI 1.20 to 5.41, $P = 0.015$), with a directionally consistent but weaker association for carotid plaque (proportional odds ratio = 1.88, 95% CI 0.96 to 3.68, $P = 0.066$) (Figure 3B). The higher CAD risk associated with a lower NLRP3 activity score is therefore mirrored in imaging-defined atherosclerotic burden.

4.4. Mediation of the coronary artery disease association by cardiometabolic risk factors

To investigate potential mechanisms underlying the higher CAD risk, we examined associations of the NLRP3 activity score with established cardiometabolic risk factors^{30,41}. A lower genetically proxied NLRP3 activity score was associated with several adverse cardiometabolic traits, including higher ApoB levels ($\beta = 0.14$, 95% CI 0.06 to 0.22, $P = 0.00078$), higher SBP ($\beta = 0.11$, 95% CI 0.05 to 0.16, $P = 0.00017$), and higher T2D risk (OR = 1.26, 95% CI 1.09 to 1.46, $P = 0.002$), but showed no association with body mass index (BMI), smoking, or alcohol consumption (Figure 3C, Supplementary Table 8). To quantify how much of the association with higher CAD risk could be attributed to these cardiometabolic pathways, we performed mediation analysis using ApoB, SBP, and T2D as mediators, with each mediator-to-CAD effect estimated in mutually adjusted MVMR (Figure 3D). Conditional F-statistics were 15.4 for ApoB, 36.7 for SBP, and 37.2 for T2D, indicating strong instruments. Together, the mediators accounted for approximately 87% of the total association between a lower genetically proxied NLRP3 activity score and higher CAD risk, with the largest indirect effect attributable to SBP (OR = 1.080, 95% CI 1.037 to 1.126) and smaller contributions from ApoB (OR = 1.047, 95% CI 1.017 to 1.079) and T2D (OR = 1.045, 95% CI 1.015 to 1.075) (Figure 3E). After accounting for all three mediators, the association between lower genetically proxied NLRP3 activity and CAD risk was attenuated and no longer statistically significant (OR = 1.026, 95% CI 0.85 to 1.24, $P = 0.79$) (Supplementary Table 9). Most of the higher CAD risk associated with a lower NLRP3 activity score was therefore attributable to adverse cardiometabolic effects, with little evidence of a residual direct effect.

4.5. Further validation of the NLRP3 activity score and robustness of the coronary artery disease association

We next examined whether the NLRP3 activity score and its association with CAD could be artefacts of the instrument rather than reflect NLRP3 biology.

To test whether the inflammatory signal at the *NLRP3* locus is driven by NLRP3 itself, we examined rare coding variants in the UK Biobank. pLOF variants in *NLRP3* were associated with significantly lower CRP ($\beta = -0.023$, 95% CI -0.032 to -0.015, $P = 9.4 \times 10^{-8}$), GlycA ($\beta = -0.018$, 95% CI -0.027 to -0.010, $P = 2.8 \times 10^{-5}$), and neutrophil counts ($\beta = -0.012$, 95% CI -0.021 to -0.0036, $P = 5.7 \times 10^{-3}$) than non-carriers (Figure 4A). Conversely, although carrier numbers were small, the three carriers of the gain-of-function variant had markedly elevated CRP and higher GlycA and neutrophil levels (Figure 4B), consistent with enhanced NLRP3 inflammasome activity. pLOF burden tests of the other protein-coding genes within ± 150 kb of *NLRP3* showed no associations with these biomarkers (Supplementary Figure 3), supporting *NLRP3* as the gene driving the inflammatory biomarker associations at this locus (Supplementary Table 10).

As a methodological positive control, we tested whether the same approach of combining *cis*-variant effects on gene expression and downstream inflammatory biomarkers into a single score could recover known drug-target biology at the *IL1RN* locus. *IL1RN* encodes interleukin-1 receptor antagonist (IL1Ra), an endogenous inhibitor of interleukin-1 signalling⁴². Recombinant IL1Ra, such as the drug anakinra, is used to treat rheumatoid arthritis (RA) and acute gout flares^{42,43}. Similar to the NLRP3 activity score, we constructed an IL1Ra activity score from variants associated with *IL1RN* expression and the downstream inflammatory biomarkers CRP, GlycA, and neutrophil count (Figure 4C). Because IL1Ra is directly measurable in plasma, we could test the score against the protein whose activity it is meant to proxy. A

higher genetically proxied IL1Ra activity score was associated with increased circulating IL1Ra protein concentration ($\beta = 4.79$, 95% CI 3.34 to 6.24, $P = 1 \times 10^{-10}$). Furthermore, consistent with the therapeutic effects of anakinra, a higher IL1Ra activity score was also associated with lower risk of RA (OR = 0.26, 95% CI 0.07 to 0.94, $P = 0.04$) and gout (OR = 0.33, 95% CI 0.23 to 0.46, $P = 3 \times 10^{-10}$) (Figure 4D, Supplementary Table 11).

To characterise the broader inflammatory pathways associated with the NLRP3 activity score, we performed proteome-wide MR. A lower genetically proxied NLRP3 activity score was associated with lower levels of 27 and higher levels of 7 circulating proteins after multiple-testing (Supplementary Table 12). Over-representation analysis of the downregulated proteins showed enrichment for acute-phase and complement-activation pathways (Figure 4E), consistent with the hepatic acute-phase response downstream of NLRP3-driven IL-1 β and IL-6 signalling. In contrast, the seven proteins raised by inhibition were enriched only for amino-acid metabolic processes, driven by three proteins (Supplementary Table 13).

Finally, the association between the NLRP3 activity score and CAD was directionally consistent across LD clumping thresholds ($r^2 = 0.1$ to $r^2 = 0.6$), in leave-one-out analyses, and in a single-variant analysis restricted to the shared colocalising variant (rs12239046) (Figure 4F, Supplementary Figure 4, Supplementary Table 14).

Collectively, these findings identify *NLRP3* as the gene underlying the inflammatory signal at this locus, and show that the CAD association is directionally consistent across sensitivity analyses.

4.6. Genetic associations of the NLRP3 activity score across diseases under investigation for NLRP3 inhibitors

Beyond CAD, we explored whether the NLRP3 activity score was associated with 15 other diseases for which NLRP3 inhibitors have been or are under clinical investigation. A lower genetically proxied NLRP3 activity score was nominally associated with lower risk of gout (OR = 0.64, 95% CI 0.47 to 0.87, $P = 0.0052$), pericarditis (OR = 0.22, 95% CI 0.06 to 0.72, $P = 0.012$), and Parkinson's disease (PD) (OR = 0.48, 95% CI 0.24 to 0.95, $P = 0.036$) (Figure 5). Among the remaining indications, a suggestive association in the same direction was observed for RA. Type 2 diabetes was the one indication where lower genetically proxied NLRP3 activity was associated with higher risk, and this estimate is reported above as a mediator of the CAD association. No clear associations were observed for the other indications examined (Supplementary Table 15). Together, these exploratory findings provide human genetic evidence consistent with potential benefit of lower NLRP3 activity in selected inflammatory indications.

5. Discussion

In this drug-target MR study, we investigated whether genetically proxied lower NLRP3 activity is associated with CAD risk. Because NLRP3 activity cannot be measured directly in large population datasets⁴⁴, we constructed a composite genetic score combining *NLRP3* expression with downstream inflammatory biomarkers. A lower genetically proxied NLRP3 activity score was associated with lower circulating IL-1 β , IL-18, and IL-6, consistent with the expected downstream effects of reduced NLRP3 activity. Despite these lower inflammatory readouts, a lower NLRP3 activity score was associated with higher CAD risk, and with greater coronary artery calcification and coronary plaque burden. Mediation analyses suggested that this association was largely accounted for by adverse cardiometabolic effects, including higher SBP, higher levels of proatherogenic lipids including ApoB, and greater T2D risk, with no evidence of a residual direct association after accounting for these pathways. In contrast, a lower NLRP3 activity score was nominally associated with lower risk of gout, pericarditis and Parkinson's disease. Overall, our data do not provide genetic support for a direct, inflammation-mediated protective effect of lifelong lower NLRP3 activity on CAD risk.

Our findings contrast with the CANTOS trial, where canakinumab reduced recurrent cardiovascular events through inhibition of IL-1 β , a downstream effector of NLRP3 inflammasome activation^{2,45}. Post hoc analyses of CANTOS showed that canakinumab lowered IL-6 but not IL-18, and that residual levels of both cytokines predicted recurrent events⁴⁶. On this basis, NLRP3 inhibition was proposed to offer greater benefit than IL-1 β blockade alone, because it would lower IL-18 in addition to IL-1 β , and direct IL-6 inhibition was proposed as an alternative strategy⁴⁶. The IL-6 arm of this hypothesis has since been tested in the ZEUS trial, which found no reduction in cardiovascular events⁵. In our analysis, a lower NLRP3 activity score was associated with lower IL-1 β , IL-18 and IL-6, yet with higher CAD risk, suggesting that the cardiovascular consequences of lower NLRP3 activity cannot be inferred from its anti-inflammatory effects alone.

A likely explanation for the higher CAD risk is the adverse cardiometabolic profile associated with lower NLRP3 activity. Mediation analyses suggested that higher SBP, ApoB levels, and T2D risk together accounted for almost all of the association with CAD. There is biological precedent linking inflammatory pathways to each of these traits. Pharmacological perturbation of inflammatory pathways, including IL-6 receptor blockade and TNF inhibition, can alter circulating lipid profiles in inflammatory disease⁴⁷. A similar pattern has been reported for genetic proxies of IL-1 α/β inhibition, which were associated with lower IL-6 and CRP but higher levels of atherogenic lipids and higher coronary heart disease risk⁴⁸. IL-18 has metabolic functions beyond inflammation, with experimental loss of IL-18 signalling impairing insulin sensitivity and energy homeostasis⁴⁹. NLRP3 has also been implicated in blood-pressure regulation through renal inflammatory mechanisms, although experimental inhibition generally lowers rather than raises blood pressure⁵⁰. The cardiometabolic effects associated with lifelong lower NLRP3 activity are therefore only partly anticipated by existing experimental and pharmacological evidence. Trials of NLRP3 inhibitors should monitor blood pressure, lipid, and glycaemic effects.

Although our findings do not support cardioprotection from lifelong lower NLRP3 activity in the general population, benefit may still exist in selected subgroups with inflammasome-driven cardiovascular risk. Clonal haematopoiesis provides one such setting, with experimental and human data implicating NLRP3-dependent IL-1 β signalling in clonal haematopoiesis-associated cardiovascular disease⁵¹. In an exploratory genomic substudy of CANTOS, participants harbouring *TET2* variants showed a greater reduction in major adverse cardiovascular events when treated with canakinumab⁵². A phase 2a trial has since evaluated the NLRP3 inhibitor DFV890 in patients with coronary heart disease and *TET2*- or *DNMT3A*-associated clonal haematopoiesis (NCT06097663).

Our findings differ from those of Schunk *et al.*, the largest previous human genetic study of the *NLRP3* locus and cardiovascular outcomes⁵³. Using a single intronic variant (rs10754555) associated with *NLRP3* expression, they reported that higher *NLRP3* activity was associated with higher CAD prevalence among people younger than 60 years of age and with higher cardiovascular mortality across primary- and secondary-prevention cohorts. The variant was associated with higher *NLRP3* expression and inflammasome activation in functional experiments, but not with lipid parameters. By contrast, our *NLRP3* activity score integrates information across eight variants that tag rs10754555 only weakly, with a maximum r^2 of 0.26 in the INTERVAL reference panel and was additionally associated with cardiometabolic traits. Because these cardiometabolic associations act on CAD in the opposite direction to the inflammatory associations, they may explain the discordance. The age- and metabolic-context dependence reported by Schunk *et al.* further suggests that the cardiovascular consequences of *NLRP3*-related genetic variation may depend on the population studied.

Although CAD was the primary focus of this study, we also examined diseases for which *NLRP3* inhibitors are in clinical development. A lower genetically proxied *NLRP3* activity score was nominally associated with lower risk of gout, pericarditis, and PD. The associations with gout and pericarditis are consistent with established *NLRP3*–IL-1 biology, with monosodium urate crystals acting as canonical *NLRP3* activators in gout and IL-1 signalling playing an established role in recurrent pericarditis^{54,55}. The PD association contrasts with a recent genetic study that used expression-based instruments for *NLRP3* inflammasome-related genes and found no evidence for a role of the *NLRP3* inflammasome in PD risk⁵⁶. Our score integrates multiple variants at the *NLRP3* locus with downstream inflammatory readouts, which may contribute to the differing findings. Overall, these exploratory disease associations are consistent with the biological relevance of the *NLRP3* activity score beyond CAD, but require replication.

We acknowledge that our study has limitations. Although cis-restriction reduces the likelihood of horizontal pleiotropy, it cannot eliminate it, because variants may influence several biological processes through mechanisms not fully captured by the intended exposure⁵⁷. The cardiometabolic associations could therefore reflect horizontal rather than vertical pleiotropy, a possibility consistent with the non-zero MR-Egger intercept. However, rare loss-of-function and gain-of-function variants in *NLRP3* affected the inflammatory biomarkers in the expected directions, and no neighbouring gene showed comparable associations, supporting *NLRP3* as the causal gene at this locus. Another limitation is that genetic variants capture small, lifelong differences in *NLRP3* activity, whereas pharmacological inhibition produces a larger perturbation over a defined treatment period. Lifelong perturbation may also be buffered by compensatory adaptation during development, and the adverse cardiometabolic associations observed here may not develop over the shorter timescale of pharmacological treatment⁵⁸. Our estimates also apply to a general population, whereas clinical studies of direct *NLRP3* inhibition in CAD have focused on patients with established coronary disease, in whom the cardiometabolic pathways implicated here are routinely treated with lipid-lowering and antihypertensive therapy. Statistical power is a further consideration. The residual direct effect on CAD, after accounting for SBP, ApoB, and T2D, was close to the null, but its confidence interval remains compatible with a modest effect in either direction. Relatedly, for several disease outcomes the available case numbers may have been insufficient to detect modest causal effects, and null findings should therefore be interpreted with caution. Our instrument is also derived from inflammatory biomarkers measured in peripheral blood and may not reflect *NLRP3* activity in disease-relevant tissues such as the coronary arteries or the brain. Establishing these tissue-specific effects will require complementary evidence, including tissue-specific molecular data and pharmacodynamic biomarker measurements in clinical studies. Finally, the *NLRP3* activity score and all disease outcome analyses were based on European-ancestry data, which limits generalisability.

In conclusion, a lower genetically proxied *NLRP3* activity score was associated with higher CAD risk, with mediation analyses suggesting that this association was largely accounted for by adverse cardiometabolic effects. These findings reflect lifelong genetic variation in the general population and do not preclude benefit from pharmacological inhibition in selected patients with inflammasome-driven cardiovascular risk. They provide no genetic support for cardioprotection from lower *NLRP3* activity, and argue for monitoring cardiometabolic effects in trials of *NLRP3* inhibitors.

6. Acknowledgements

This research has been conducted using the UK Biobank Resource under Application Number 608471. We gratefully acknowledge All of Us participants for their contributions, without whom this research would not have been possible. We also thank the National Institutes of Health's All of Us Research Program for making available the participant data examined in this study. We want to acknowledge the participants and investigators of the FinnGen study. We thank the participants and investigators of UK Biobank, INTERVAL, the Million Veteran Program, SCAPIS and deCODE genetics, and all consortia that made their summary statistics publicly available. This work was performed using resources provided by the Cambridge Service for Data Driven Discovery (CSD3) operated by the University of Cambridge Research Computing Service (www.csd3.cam.ac.uk), provided by Dell EMC and Intel using Tier-2 funding from the Engineering and Physical Sciences Research Council (capital grant EP/T022159/1), and DiRAC funding from the Science and Technology Facilities Council (www.dirac.ac.uk).

7. Funding

This work was funded by ...

8. Disclosure of interest

N.H. holds a Non-Clinical PhD Studentship funded by AstraZeneca. S.B. is an employee of Sequoia Genetics. D.S.P. was an employee and shareholder of AstraZeneca during the duration of this work.

9. Data availability

All GWAS summary statistics analysed in this study are publicly available from the sources listed in Supplementary Tables 1–4. Individual-level data from UK Biobank and the other contributing studies are available to approved researchers through their respective access procedures. This study used data from the All of Us Research Program, available to authorized users on the Researcher Workbench. The code used for all analyses and figures is available at https://github.com/nickhir/NLRP3_MR_manuscript.

10. Ethical Approval

This study used summary statistics from published genome-wide association studies, each of which obtained ethical approval and participants' informed consent as described in the original publications; no further approval was required for their use. Individual-level UK Biobank data were analysed under Application Number 608471. UK Biobank has Research Tissue Bank approval from the North West Multi-centre Research Ethics Committee (11/NW/0382, renewed as 16/NW/0274 and 21/NW/0157), and all participants gave written informed consent. The INTERVAL study was approved by the National Research Ethics Service (11/EE/0538), and all participants gave informed consent. All of Us participants provided informed consent, and analyses of de-identified data within the All of Us Researcher Workbench are not subject to separate institutional review board review. The study was conducted in accordance with the Declaration of Helsinki.

515 **11. Pre-registered Clinical Trial Number**

516 Not applicable.

12. Figures

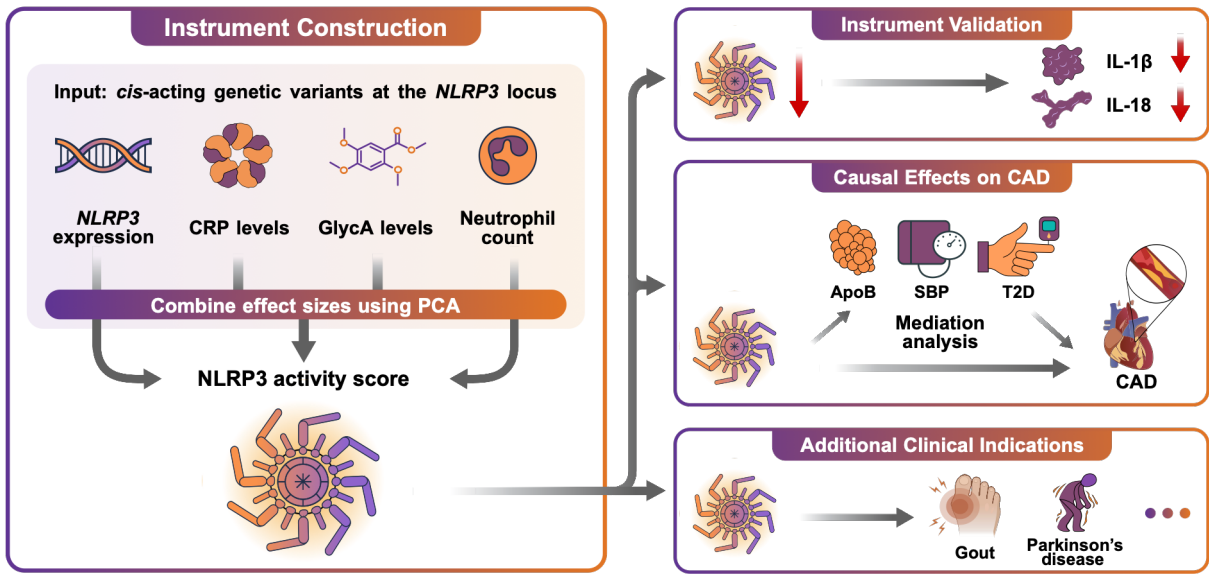


Figure 1. The study design. A genetic score for NLRP3 activity was constructed from variants at the *NLRP3* locus associated with *NLRP3* expression, CRP, GlycA, and neutrophil count, whose effect sizes were combined by PCA into a single NLRP3 activity score. The score was validated against the canonical inflammasome effectors IL-1 β and IL-18. Using Mendelian randomization, the score was applied to estimate the effect of genetically proxied NLRP3 activity on CAD risk and, through mediation analysis, to quantify the extent to which this effect is mediated by cardiometabolic risk factors, namely ApoB, SBP, and T2D. The score was further applied across additional clinical indications, including gout, pericarditis, and Parkinson's disease.

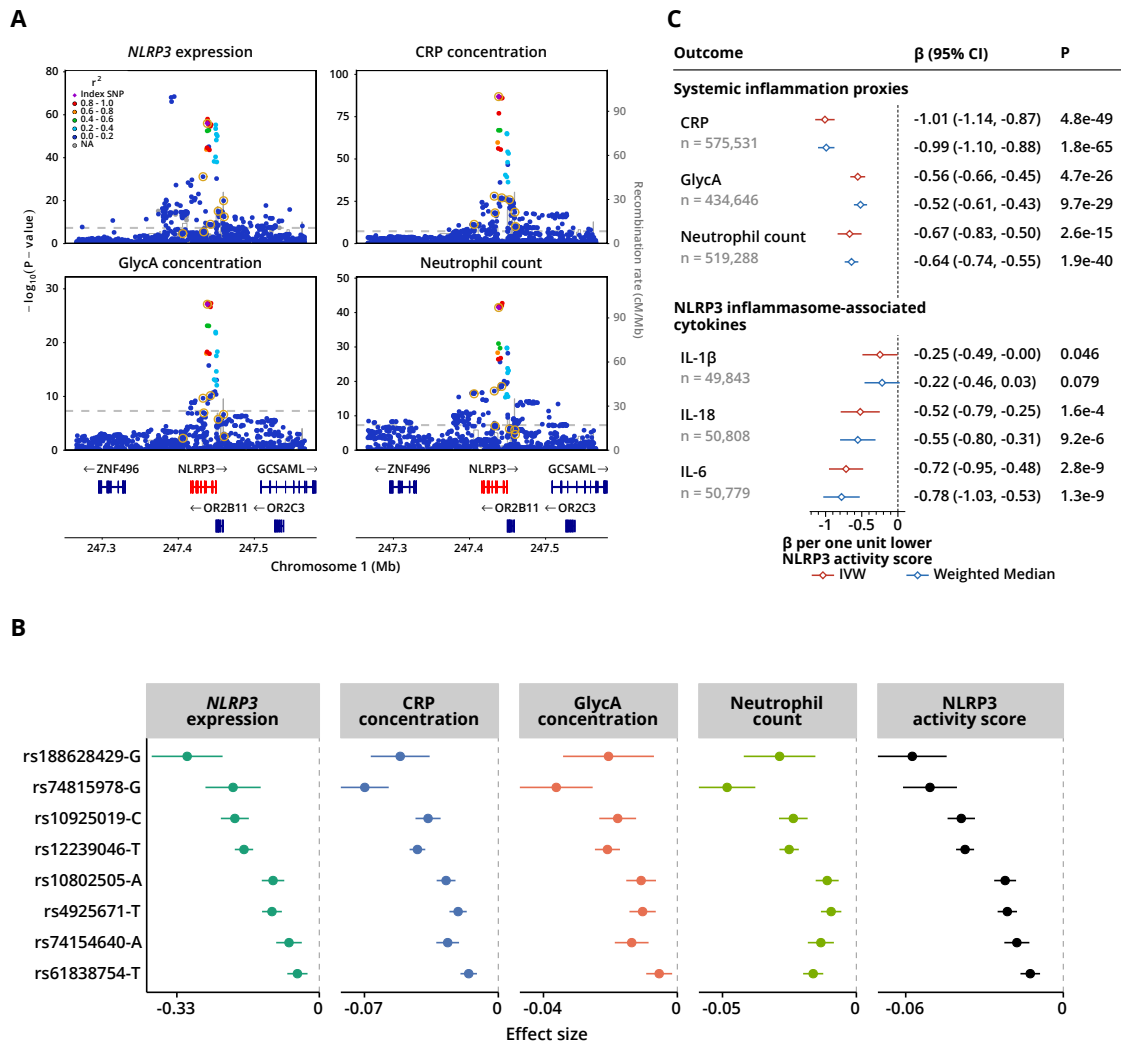


Figure 2. Construction and validation of the genetic score for NLRP3 activity. (A) Regional association plots for *NLRP3* expression, CRP concentration, GlycA concentration, and neutrophil count at the *NLRP3* locus. The eight variants taken forward as instruments are circled in gold. (B) Per-allele effect sizes of each variant on individual traits and on the NLRP3 activity score. The latter represents the PC1-weighted composite of effect sizes across the four traits. The effect allele, shown after each rsID, is the allele that lowers the NLRP3 activity score. (C) Forest plot of MR estimates for the effect of the NLRP3 activity score on the inflammatory biomarkers used in its construction (CRP, GlycA, and neutrophil count) and on cytokines not used in its construction (IL-1 β , IL-18, and IL-6). Estimates are reported as SD changes per one-unit lower NLRP3 activity score, from inverse-variance weighted and weighted median MR. PC1, principal component.

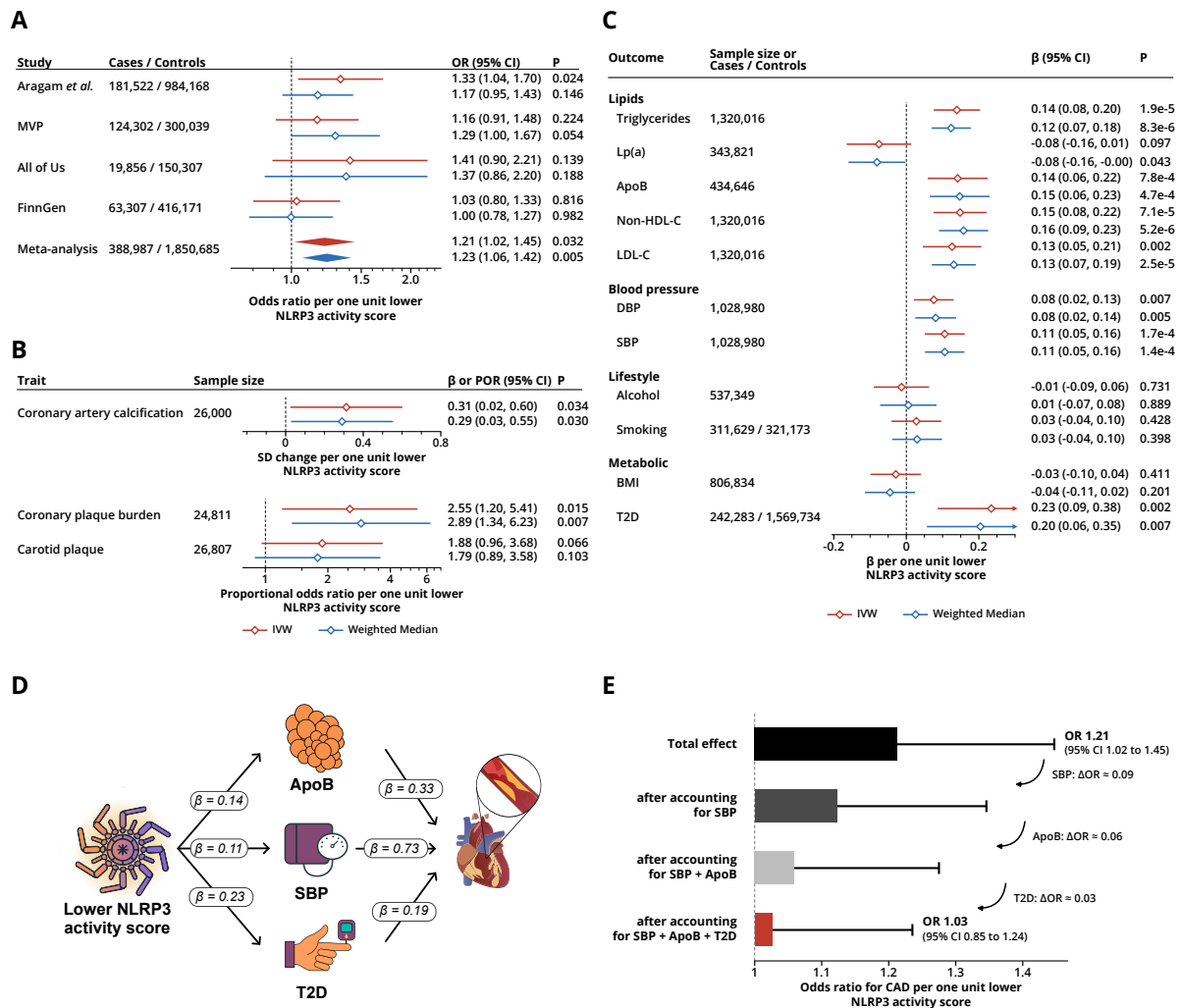


Figure 3. Association of the NLRP3 activity score with coronary artery disease risk and its mediation by cardiometabolic risk factors. (A) Forest plot of MR estimates for the effect of a lower NLRP3 activity score on CAD risk in each contributing GWAS (Aragam *et al.*, MVP, All of Us, and FinnGen) and in their fixed-effect meta-analysis. (B) MR estimates for the association of a lower NLRP3 activity score with coronary artery calcification, coronary plaque burden, and carotid plaque. Coronary artery calcification is reported as SD change, whereas coronary and carotid plaque outcomes are reported as proportional odds ratios. (C) Forest plot of MR estimates for the effect of a lower NLRP3 activity score on cardiometabolic risk factors, grouped into lipid, blood pressure, lifestyle, and metabolic traits. Estimates for continuous outcomes are reported in SD units and estimates for binary outcomes as log odds ratios. (D) Mediation diagram of the pathways from a lower NLRP3 activity score to CAD through ApoB, SBP, and T2D, with each mediator-to-CAD effect estimated in mutually adjusted MVMR. Effects of the score on ApoB and SBP are in SD units and on T2D in log odds. Mediator-to-CAD effects are log odds ratios for CAD per SD higher ApoB or SBP, and per one-unit higher log odds of T2D liability. (E) Waterfall plot of the progressive attenuation of the total effect of a lower NLRP3 activity score on CAD risk as SBP, ApoB, and T2D are sequentially accounted for, resulting in the residual direct effect. DBP, diastolic blood pressure; Lp(a), lipoprotein(a); Non-HDL-C, non-high-density lipoprotein cholesterol; POR, proportional odds ratio.

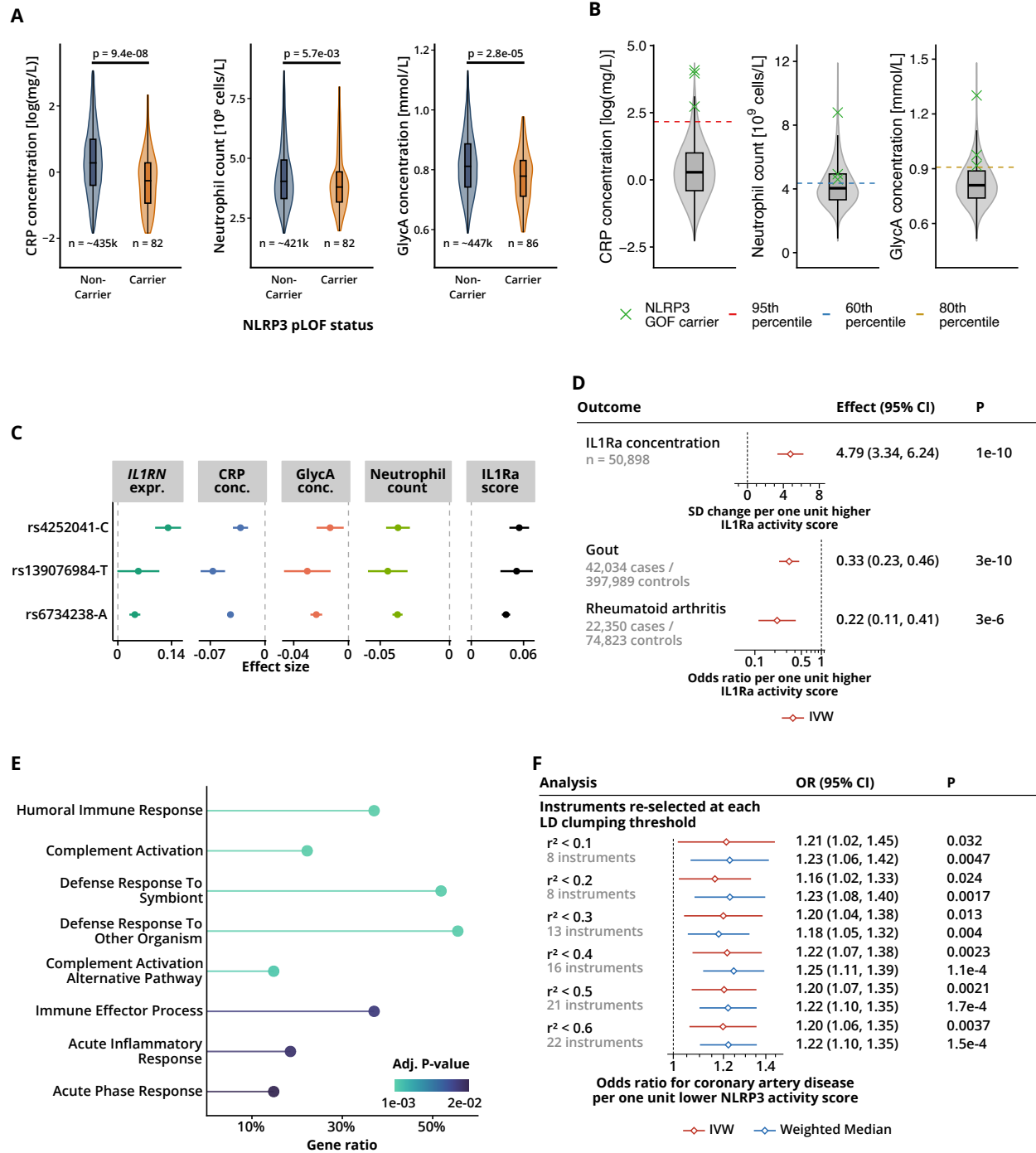
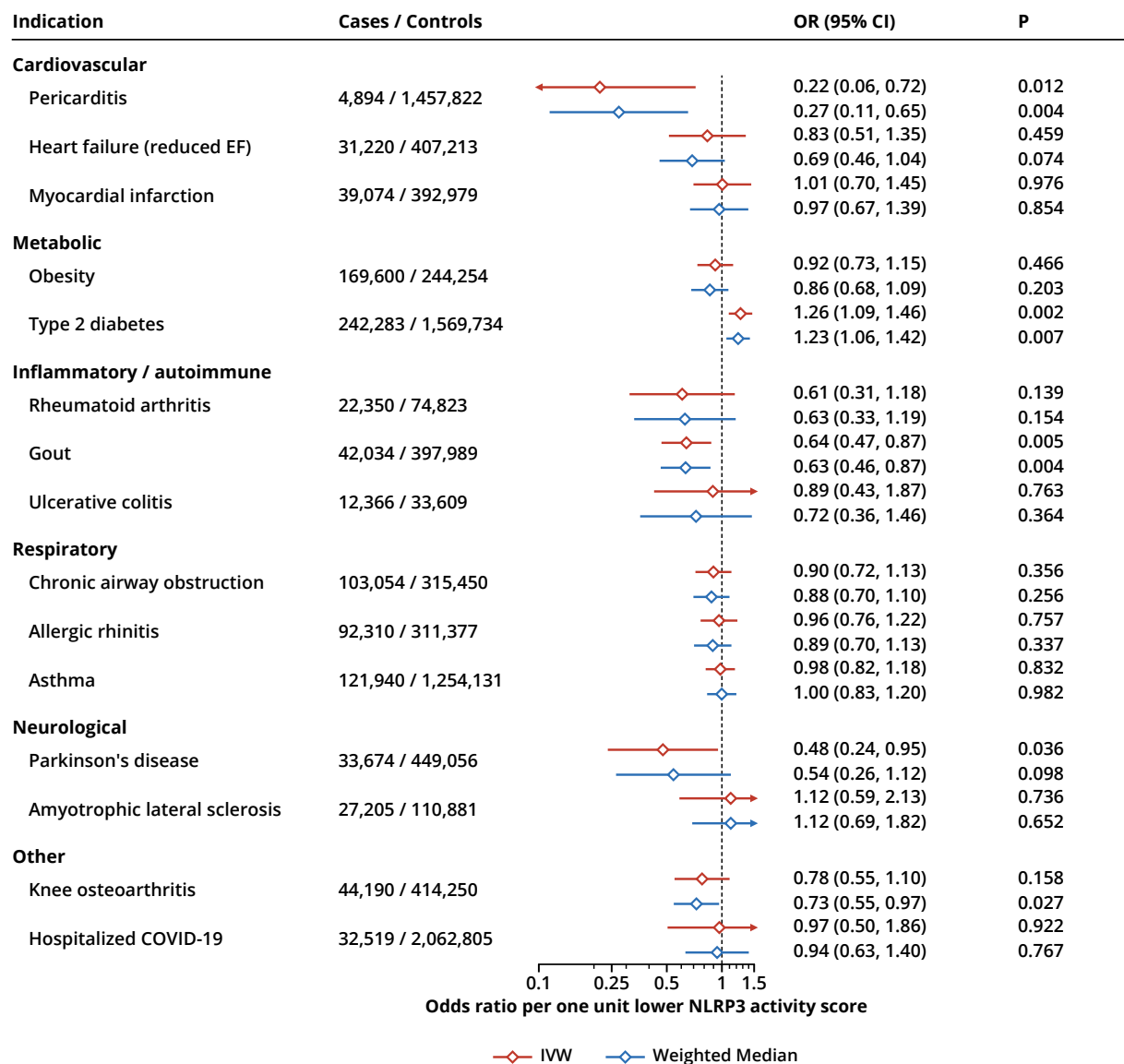


Figure 4. Supporting analyses for the *NLRP3* activity score and robustness of the coronary artery disease association. (A) Violin plots comparing circulating CRP, neutrophil count, and GlycA levels between carriers and non-carriers of predicted loss-of-function variants in *NLRP3*. The displayed distributions exclude observations outside the 1st and 99th percentiles for visualisation. Inset boxplots show the median and interquartile range, with whiskers extending to the most extreme displayed observations within 1.5 times the interquartile range. *P* values are from SAIGE-GENE+ gene-based burden tests. (B) Distribution of circulating CRP, neutrophil count, and GlycA levels in UK Biobank participants, with carriers of the *NLRP3* gain-of-function variant highlighted. Dashed lines indicate reference percentiles. (C)

Per-allele effect sizes of variants at the *IL1RN* locus on *IL1RN* expression, CRP, GlycA, neutrophil count, and the genetically proxied IL1Ra activity score. **(D)** Forest plot of MR estimates for the effect of a higher IL1Ra activity score on circulating IL1Ra protein concentration, gout, and rheumatoid arthritis. The protein estimate is reported as SD change per one-unit higher score and disease estimates as odds ratios. Estimates are from random-effects IVW MR only, as only three variants were available, making weighted-median analysis an uninformative sensitivity analysis. **(E)** Over-representation analysis of circulating proteins decreased with a lower genetically proxied NLRP3 activity score in proteome-wide MR. Gene Ontology Biological Process terms with adjusted $P < 0.05$ are shown. The x-axis shows the gene ratio, calculated as the proportion of significant input proteins annotated to each GO term. Colour indicates the adjusted P value. **(F)** Forest plot of MR estimates for the effect of a lower NLRP3 activity score on CAD risk across alternative LD clumping thresholds, with instruments re-selected at each threshold. The number of instruments at each threshold is shown in the figure. GO, Gene Ontology; GOF, gain-of-function.

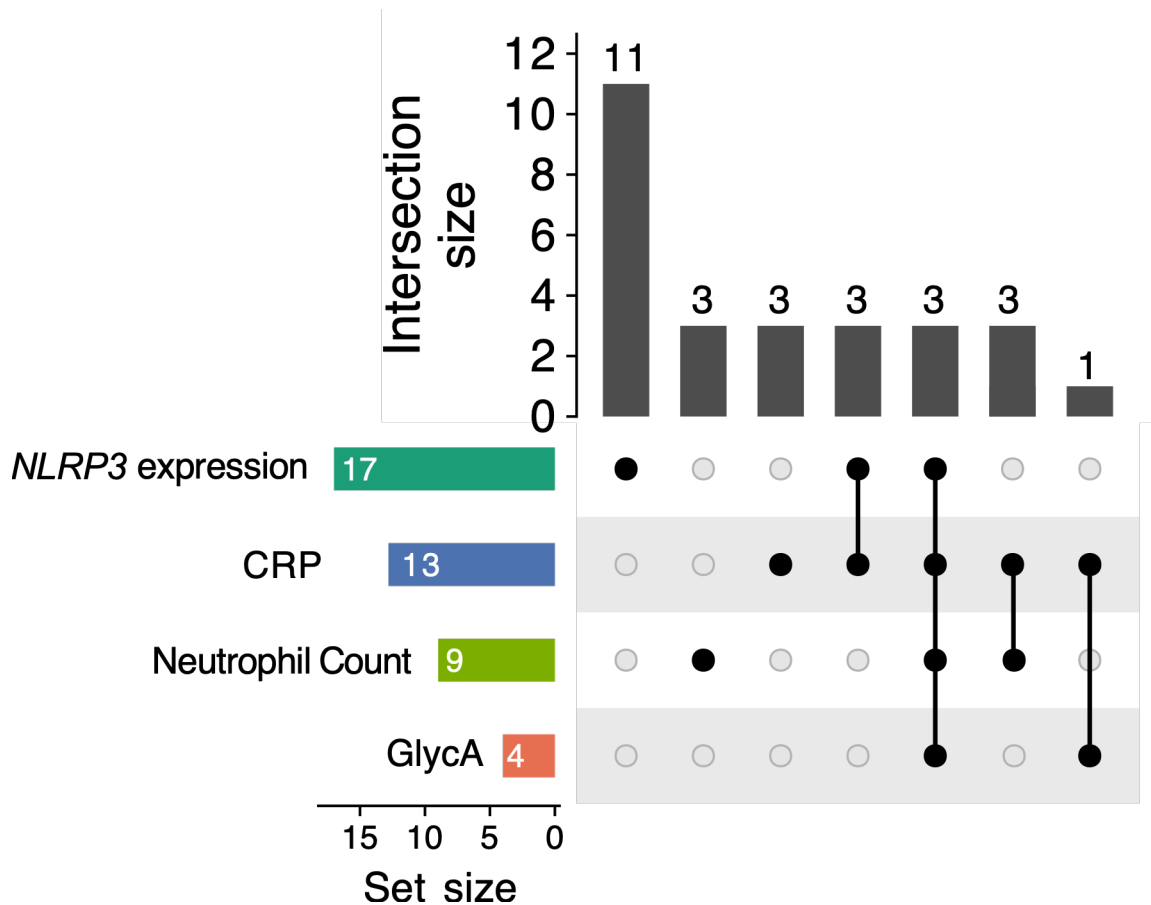


577

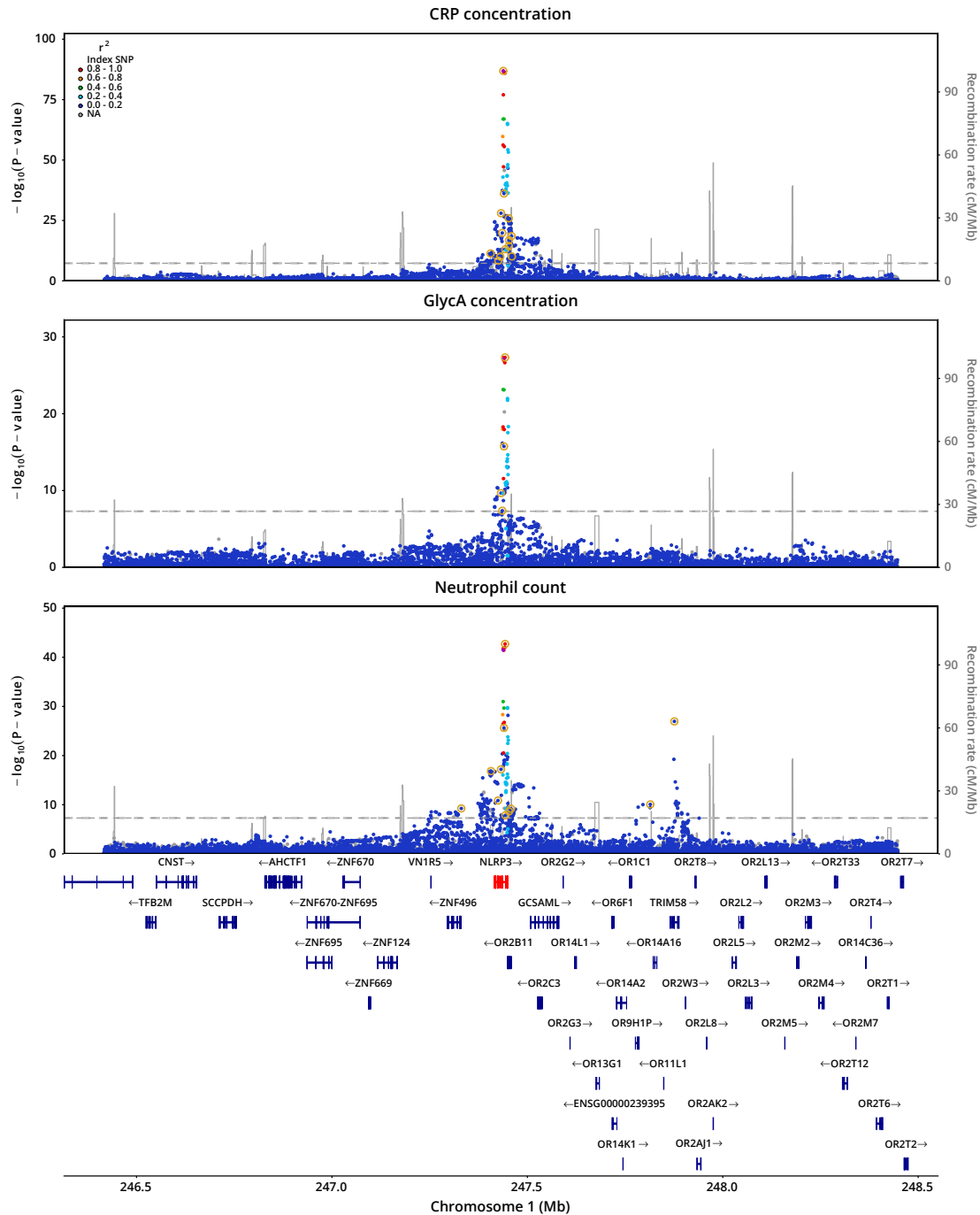
Figure 5. NLRP3 activity score associations across diseases under investigation for NLRP3 inhibition.

Forest plot showing MR estimates for the association of a lower NLRP3 activity score with diseases for which NLRP3 inhibitors have been or are being evaluated in clinical trials. Outcomes are grouped by disease area. Estimates are shown as odds ratios per one-unit lower NLRP3 activity score, from inverse-variance weighted and weighted median MR. Parkinson's disease cases comprise 15,056 diagnosed cases and 18,618 UK Biobank proxy cases. EF, ejection fraction.

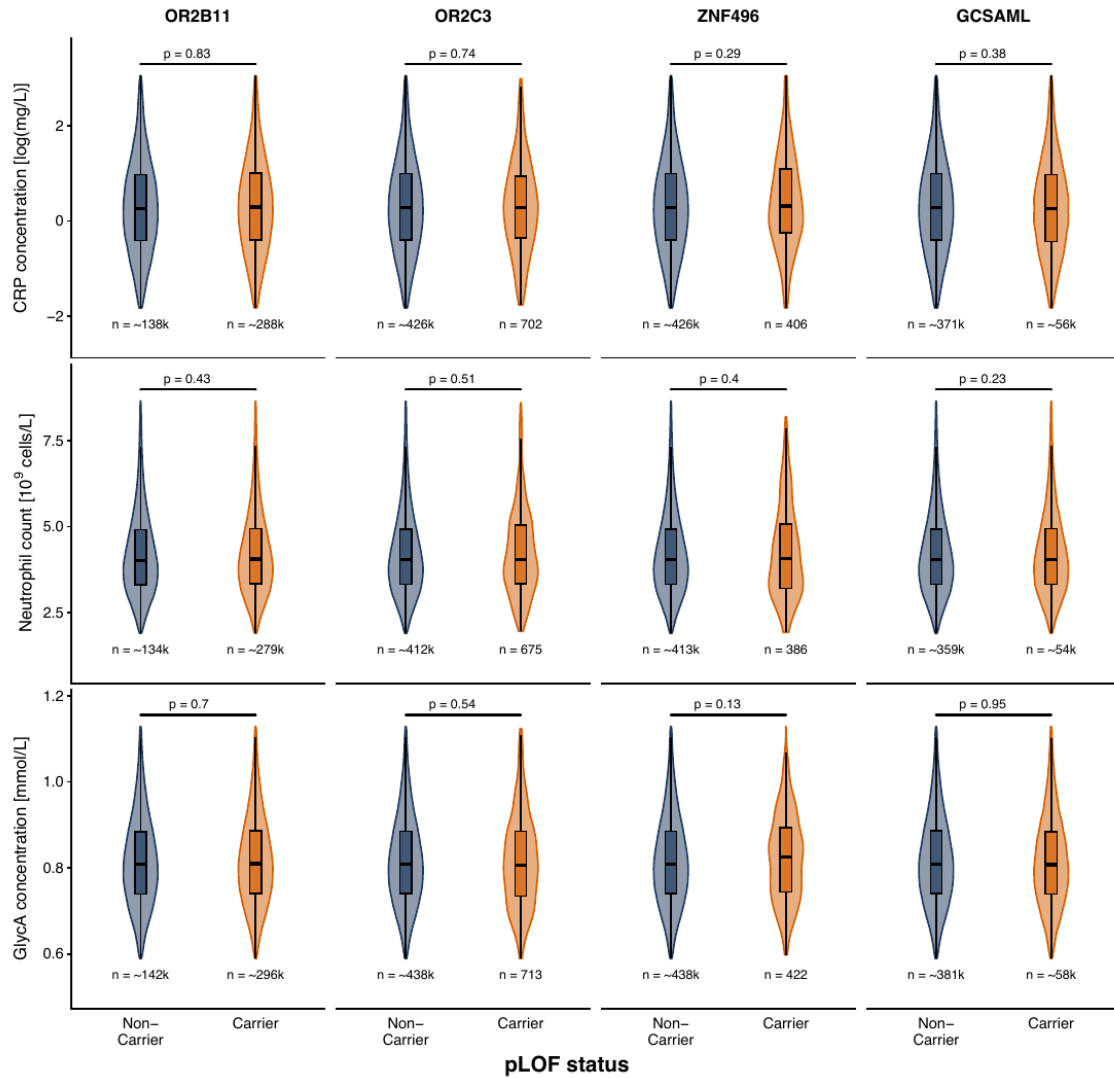
13. Supplementary Data



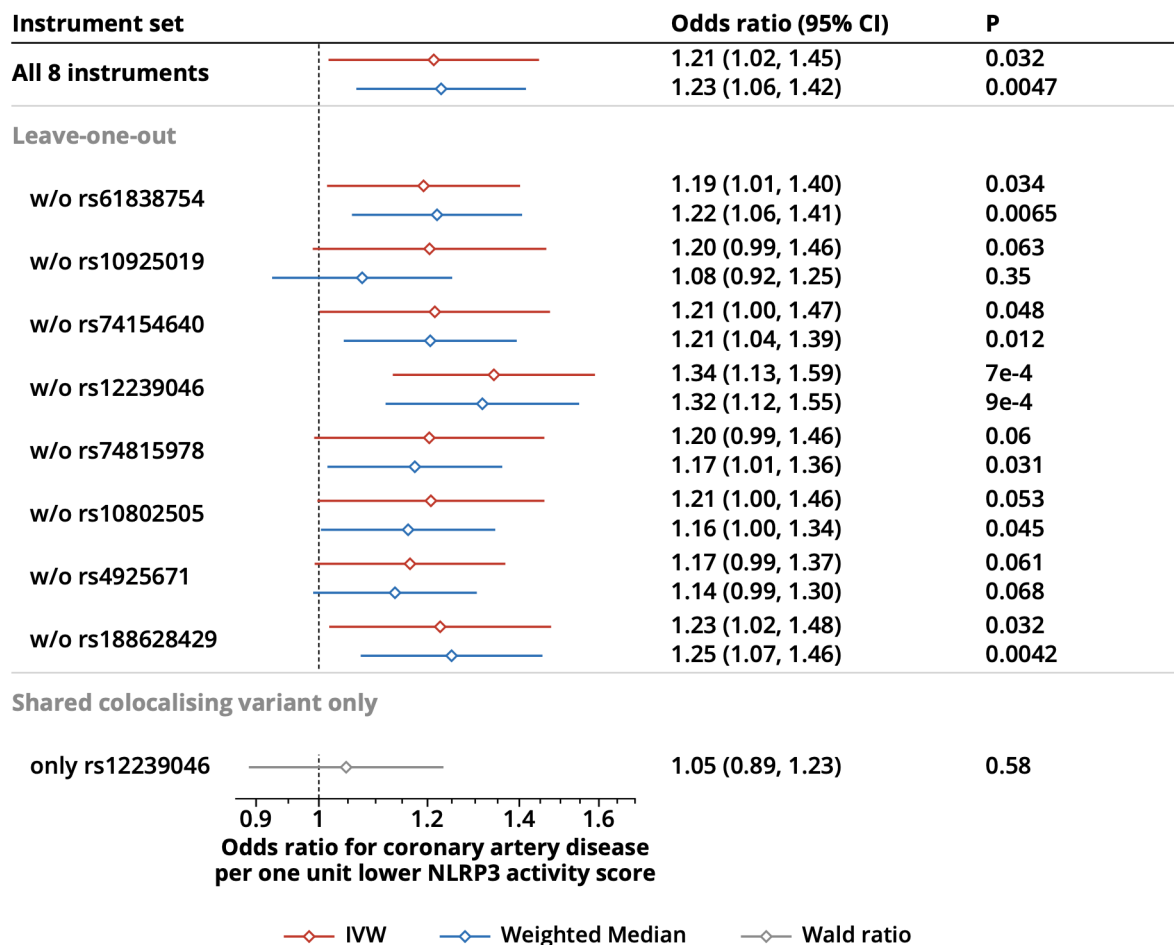
Supplementary Figure 1. Overlap of variants associated with the four traits used to construct the NLRP3 activity score. UpSet plot showing the overlap of variants within ± 150 kb of NLRP3 associated at genome-wide significance with whole-blood NLRP3 expression, circulating CRP and GlycA levels, or blood neutrophil count after LD clumping at $r^2 < 0.1$. Set size indicates the number of independent genome-wide significant variants per trait, and intersection size indicates the number of variants belonging to each specific combination of traits. Variants associated with at least two traits and with effect estimates available for all four were retained, yielding the eight variants used to construct the NLRP3 activity score. CRP, C-reactive protein; GlycA, glycoprotein acetylation; LD, linkage disequilibrium.



Supplementary Figure 2. Regional association plots for inflammatory traits within ± 1 Mb of the *NLRP3* locus. Regional association plots for CRP concentration, GlycA concentration, and neutrophil count within a ± 1 Mb window surrounding *NLRP3*. Variants highlighted with gold circles are approximately independent genome-wide significant variants identified after LD clumping at $r^2 < 0.1$. Within this region, the shared association signal across traits was centred on the *NLRP3* locus. Additional neutrophil count association signals were not accompanied by corresponding associations with CRP or GlycA. CRP, C-reactive protein; GlycA, glycoprotein acetylation; LD, linkage disequilibrium.



Supplementary Figure 3. pLOF burden analysis of protein-coding genes surrounding *NLRP3*. Violin plots comparing circulating CRP, neutrophil count, and GlycA levels between carriers and non-carriers of predicted loss-of-function (pLOF) variants in all protein-coding genes within ± 150 kb of *NLRP3*. The displayed distributions exclude observations outside the 1st and 99th percentiles for visualisation. Inset boxplots show the median and interquartile range, with whiskers extending to the most extreme displayed observations within 1.5 times the interquartile range. P values are from SAIGE-GENE+ gene-based burden tests. CRP, C-reactive protein; GlycA, glycoprotein acetylation; pLOF, predicted loss-of-function.



Supplementary Figure 4. Sensitivity analyses for the association of the NLRP3 activity score with coronary artery disease risk. Forest plot showing MR estimates for the effect of a one-unit lower NLRP3 activity score on CAD risk using the full eight-variant instrument set, sequential leave-one-out analyses excluding each variant in turn, and a single-variant analysis restricted to the shared colocalising variant rs12239046. Estimates for the full and leave-one-out instrument sets are shown using inverse-variance weighted and weighted median MR; the rs12239046-only estimate is shown as a Wald ratio. Odds ratios and 95% confidence intervals are based on the fixed-effect meta-analysis of the four contributing CAD GWAS. CAD, coronary artery disease; CI, confidence interval; GWAS, genome-wide association study; IVW, inverse-variance weighted; MR, Mendelian randomization.

Supplementary Information

14.1. Construction of the NLRP3 and IL1Ra activity scores

To construct the NLRP3 activity score, we used genetic associations for four traits related to NLRP3 inflammasome activation: whole-blood *NLRP3* expression, circulating CRP and GlycA levels, and blood neutrophil count. For each trait, variants within ± 150 kb of *NLRP3* were identified and LD-clumped using PLINK2⁵⁹ with a 250 kb clumping window, a genome-wide significance threshold of $P < 5 \times 10^{-8}$, and a primary clumping threshold of $r^2 < 0.1$. The LD reference panel comprised 11,863 European-ancestry whole-genome sequences from INTERVAL. Variants whose effect-allele frequency in the panel differed from the GWAS frequency by more than 0.05 were excluded from the reference panel. Because standard clumping may select different lead variants for different traits even when these tag the same underlying signal, we next identified variants in high LD with each trait-specific lead variant using $r^2 > 0.95$ within a 40 kb window. These high-LD variant sets were treated as signal blocks, and blocks were merged across traits if they shared at least one variant. Components represented in at least two of the four traits were retained for score construction, of which there were ten. For each retained component, we selected a single representative variant, prioritising variants available across all four datasets and then choosing the variant with the lowest CRP association P -value. Where the representative variant was absent from one or more datasets, we substituted the variant in highest LD with it within 50 kb, provided r^2 was at least 0.9. Two representatives were still missing from at least one dataset after this substitution and were excluded, leaving eight variants for construction of the NLRP3 activity score (Supplementary Figure 1). We further required instruments to be specific to the target gene. Using INTERVAL whole-blood eQTL summary statistics, a variant was excluded if it was associated ($P < 1 \times 10^{-3}$) with expression of another gene tested in *cis*, and this association persisted ($P < 1 \times 10^{-3}$) after conditioning on that gene's lead eQTL. This increased the specificity of the instruments for their target gene.

For each retained variant j , trait-specific genetic effect estimates were assembled into the vector $\hat{\beta}_j = (\hat{\beta}_{j,1}, \dots, \hat{\beta}_{j,K})^T$, where $K = 4$ denotes the four traits used to construct the score. We centred and scaled the trait-specific effect estimates to have mean zero and unit variance across variants before applying PCA to the resulting variant-by-trait matrix. Loadings from the first principal component were extracted and used as fixed weights $w = (w_1, \dots, w_K)^T$ to combine trait-specific variant effects into a single latent factor:

$$\tilde{\beta}_j = w^T \hat{\beta}_j = \sum_{k=1}^4 w_k \hat{\beta}_{j,k}$$

This latent factor was defined as the NLRP3 activity score. Because its raw scale is not directly interpretable, we scaled it to CRP. Specifically, we estimated a scaling constant c from a weighted linear regression of the CRP effect estimate $\hat{\beta}_{j,CRP}$ on the unscaled score effect $\tilde{\beta}_j$, with the intercept constrained to zero and inverse-variance weights based on the CRP standard errors. The scaled variant effect on the NLRP3 activity score was then defined as:

$$\hat{\beta}_j^{NLRP3 \text{ score}} = c \tilde{\beta}_j = c \sum_{k=1}^4 w_k \hat{\beta}_{j,k}$$

The score was oriented so that lower values corresponded to lower CRP. By construction, a one-unit decrease in the scaled score corresponds approximately to a one standard deviation decrease in CRP.

Standard errors for $\hat{\beta}_j^{NLRP3 \text{ score}}$ were estimated using the delta method²⁴. For each variant, we defined a variance-covariance matrix for the trait-specific effect estimates. Because the CRP, GlycA, and neutrophil count GWAS may include overlapping participants, we used a conservative approach and assumed perfectly correlated sampling errors across these biomarker traits. This assumption inflates standard errors and therefore yields conservative inference. As *NLRP3* expression was measured in a distinct cohort,

expression quantitative trait loci effect estimates were assumed to be independent of the biomarker effect estimates.

The same procedure was applied at the *IL1RN* locus to construct a genetically proxied interleukin-1 receptor antagonist (IL1Ra) activity score, using *IL1RN* expression and the same blood-based inflammatory traits.

14.2. *NLRP3* rare variant phenotype processing

For *NLRP3* pLOF burden analyses and gain-of-function carrier analyses, CRP levels and GlycA concentrations were adjusted for age, age², sex, age × sex, age² × sex, assessment centre, genotyping array, and the first 20 genetic principal components before inverse-rank normalisation. GlycA concentrations were additionally adjusted for technical variation using the ukbmmr package (version 3.3.1)⁶⁰. Blood neutrophil count was adjusted following the blood-cell trait processing framework described by Astle *et al.*, including adjustment for technical variation, age- and sex-related effects, height and weight, smoking, alcohol-related covariates, recruitment centre, and genetic principal components before inverse-rank normalisation⁶¹.

14.3. Proteome-wide multiple-testing correction and over-representation analysis

To account for correlation between protein measurements, we estimated the effective number of independent protein tests. To do this, protein measurements were first adjusted for age, sex, age × sex, assessment centre, genotyping batch, Olink batch, and the first 20 genetic principal components. Pairwise Pearson correlations were then calculated across adjusted protein measurements, and eigenvalue decomposition of the resulting correlation matrix was used to estimate the effective number of independent tests. The effective number of tests was defined as the number of principal components required to explain 95% of the total variance⁴⁰. This yielded 1,821 effective tests, corresponding to a significance threshold of $P < 2.7 \times 10^{-5}$.

To assess whether proteins altered by lower genetically proxied NLRP3 activity were enriched for specific biological processes, we performed over-representation analysis using clusterProfiler (version 4.10.1)⁶². Proteins whose circulating levels were significantly lower or higher with lower genetically proxied NLRP3 activity and passed the multiple-testing threshold were tested separately for enrichment against Gene Ontology Biological Process terms obtained using msigdb (version 25.1.1)⁶³. The background set comprised all 2,922 tested proteins. *P* values were adjusted using the Benjamini-Hochberg method⁶⁴.

14.4. Selection of disease indications from clinical trial registries

To identify disease indications for which selective NLRP3 inhibitors have entered clinical evaluation, we searched ClinicalTrials.gov, the EU Clinical Trials Register (EudraCT), the EU Clinical Trials Information System (CTIS) and ISRCTN on the 28th of April 2026 using two complementary strategies.

Search 1 (target-level search): Each registry was searched for the term "NLRP3", restricted to interventional studies.

Search 2 (drug-name search): Each registry was searched for the following compound names: dapansutrile, OLT1177, DFV890, IFM-2427, selnoflast, RO7486967, RG6418, inzomelid, IZD174, somalix, IZD334, emlenoflast, NT-0796, ruvonoflast, NT-0249, VTX2735, VTX3232, VENT-02, ZYIL1, usnoflast, ACI-19764, AZD4144, HL-400, JTE-162, JT002, GDC-2394, and MCC950.

Results from both searches were merged and de-duplicated, with trials registered in more than one registry counted once. Each record was screened and a trial was included only if the intervention was a compound designed specifically to inhibit NLRP3. In addition, rheumatoid arthritis was included because a selective NLRP3 inhibitor (MCC950/CP-456773) was previously evaluated in a Phase II trial that was terminated due to off-target hepatotoxicity rather than lack of efficacy⁶⁵. The complete list of eligible trials and the disease indications extracted from them are provided in Supplementary Table 16.

References

1. Ridker PM. Clinician's Guide to Reducing Inflammation to Reduce Atherothrombotic Risk: JACC Review Topic of the Week. *J Am Coll Cardiol* 2018;**72**:3320–3331.
2. Ridker PM, Everett BM, Thuren T, MacFadyen JG, Chang WH, Ballantyne C, et al. Antiinflammatory Therapy with Canakinumab for Atherosclerotic Disease. *N Engl J Med* 2017;**377**:1119–1131.
3. Nidorf SM, Fiolet ATL, Mosterd A, Eikelboom JW, Schut A, Opstal TSJ, et al. Colchicine in Patients with Chronic Coronary Disease. *N Engl J Med* 2020;**383**:1838–1847.
4. Jolly SS, d'Entremont M-A, Lee SF, Mian R, Tyrwhitt J, Kedev S, et al. Colchicine in Acute Myocardial Infarction. *N Engl J Med* 2025;**392**:633–642.
5. Yao Z, Blaha MJ, Zahid S. Preparing for the Full ZEUS Results and the Unfinished Question of IL-6 Inhibition. *Am J Prev Cardiol* 2026:101799.
6. Swanson KV, Deng M, Ting JP-Y. The NLRP3 inflammasome: molecular activation and regulation to therapeutics. *Nat Rev Immunol* 2019;**19**:477–489.
7. Lee J-K, Kim S-H, Lewis EC, Azam T, Reznikov LL, Dinarello CA. Differences in signaling pathways by IL-1 β and IL-18. *Proc Natl Acad Sci U S A* 2004;**101**:8815–8820.
8. Bartlett DB, Connelly MA, AbouAssi H, Bateman LA, Tune KN, Huebner JL, et al. A novel inflammatory biomarker, GlycA, associates with disease activity in rheumatoid arthritis and cardiovascular risk in BMI-matched controls. *Arthritis Res Ther* 2016;**18**:86.
9. Sreejit G, Nooti SK, Jaggers RM, Athmanathan B, Ho Park K, Al-Sharea A, et al. Retention of the NLRP3 Inflammasome-Primed Neutrophils in the Bone Marrow Is Essential for Myocardial Infarction-Induced Granulopoiesis. *Circulation* 2022;**145**:31–44.
10. van der Heijden T, Kritikou E, Venema W, van Duijn J, van Santbrink PJ, Slütter B, et al. NLRP3 Inflammasome Inhibition by MCC950 Reduces Atherosclerotic Lesion Development in Apolipoprotein E-Deficient Mice—Brief Report. *Arterioscler Thromb Vasc Biol* 2017;**37**:1457–1461.
11. Jiang Y, Qiang Z, Liu Y, Zhu L, Xiao L, Du Z, et al. Diverse functions of NLRP3 inflammasome in PANoptosis and diseases. *Cell Death Discov* 2025;**11**:389.
12. Ference BA, Robinson JG, Brook RD, Catapano AL, Chapman MJ, Neff DR, et al. Variation in PCSK9 and HMGCR and Risk of Cardiovascular Disease and Diabetes. *N Engl J Med* 2016;**375**:2144–2153.
13. Gregson JM, Freitag DF, Surendran P, Stitzel NO, Chowdhury R, Burgess S, et al. Genetic invalidation of Lp-PLA2 as a therapeutic target: Large-scale study of five functional Lp-PLA2-lowering alleles. *Eur J Prev Cardiol* 2017;**24**:492–504.
14. Said S, Pazoki R, Karhunen V, Vösa U, Ligthart S, Bodinier B, et al. Genetic analysis of over half a million people characterises C-reactive protein loci. *Nat Commun* 2022;**13**:2198.

15. Zoodma M, Beuchel C, Yasmeen S, Kohleick L, Nepal A, Koprulu M, et al. A genetic map of human metabolism across the allele frequency spectrum. *Nat Genet* 2025;**57**:2445–2455.
16. Chen M-H, Raffield LM, Mousas A, Sakaue S, Huffman JE, Moscati A, et al. Trans-ethnic and Ancestry-Specific Blood-Cell Genetics in 746,667 Individuals from 5 Global Populations. *Cell* 2020;**182**:1198-1213.e14.
17. Tokolyi A, Persyn E, Nath AP, Burnham KL, Marten J, Vanderstichele T, et al. The contribution of genetic determinants of blood gene expression and splicing to molecular phenotypes and health outcomes. *Nat Genet* 2025;**57**:616–625.
18. Sun BB, Chiou J, Traylor M, Benner C, Hsu Y-H, Richardson TG, et al. Plasma proteomic associations with genetics and health in the UK Biobank. *Nature* 2023;**622**:329–338.
19. Aragam KG, Jiang T, Goel A, Kanoni S, Wolford BN, Atri DS, et al. Discovery and systematic characterization of risk variants and genes for coronary artery disease in over a million participants. *Nat Genet* 2022;**54**:1803–1815.
20. Verma A, Huffman JE, Rodriguez A, Conery M, Liu M, Ho Y-L, et al. Diversity and scale: Genetic architecture of 2068 traits in the VA Million Veteran Program. *Science* 2024;**385**:eadj1182.
21. Lu W, Carroll RJ, Solomonson M, Guez J, He MK, Marten DJ, et al. Systematic common and rare variant association testing in 392,030 whole genomes in All of Us. *medRxiv* 2026. Available from: <https://doi.org/10.64898/2026.05.08.26350964>.
22. Kurki MI, Karjalainen J, Palta P, Sipilä TP, Kristiansson K, Donner KM, et al. FinnGen provides genetic insights from a well-phenotyped isolated population. *Nature* 2023;**613**:508–518.
23. Gummeson A, Lundmark P, Chen QS, Björnson E, Dekkers KF, Hammar U, et al. A genome-wide association study of imaging-defined atherosclerosis. *Nat Commun* 2025;**16**:2266.
24. Rietveld CA, de Vlaming R, Slob EAW. The identification of mediating effects using genome-based restricted maximum likelihood estimation. *PLOS Genet* 2023;**19**:e1010638.
25. Foley CN, Staley JR, Breen PG, Sun BB, Kirk PDW, Burgess S, et al. A fast and efficient colocalization algorithm for identifying shared genetic risk factors across multiple traits. *Nat Commun* 2021;**12**:764.
26. Yang J, Ferreira T, Morris AP, Medland SE, Madden PAF, Heath AC, et al. Conditional and joint multiple-SNP analysis of GWAS summary statistics identifies additional variants influencing complex traits. *Nat Genet* 2012;**44**:369–375.
27. Burgess S, Davey Smith G, Davies NM, Dudbridge F, Gill D, Glymour MM, et al. Guidelines for performing Mendelian randomization investigations: update for summer 2023. *Wellcome Open Res* 2023;**4**:186.
28. Burgess S, Yavorska O. *MendelianRandomization: Mendelian Randomization Package*. 2024.
29. Sanderson E, Davey Smith G, Windmeijer F, Bowden J. An examination of multivariable Mendelian randomization in the single-sample and two-sample summary data settings. *Int J Epidemiol* 2019;**48**:713–727.

30. Libby P. The changing landscape of atherosclerosis. *Nature* 2021;**592**:524–533.
31. Marston NA, Giugliano RP, Melloni GEM, Park J-G, Morrill V, Blazing MA, et al. Association of Apolipoprotein B-Containing Lipoproteins and Risk of Myocardial Infarction in Individuals With and Without Atherosclerosis: Distinguishing Between Particle Concentration, Type, and Content. *JAMA Cardiol* 2022;**7**:250–256.
32. VanderWeele TJ, Vansteelandt S. Mediation Analysis with Multiple Mediators. *Epidemiol Methods* 2014;**2**:95–115.
33. Suzuki K, Hatzikotoulas K, Southam L, Taylor HJ, Yin X, Lorenz KM, et al. Genetic drivers of heterogeneity in type 2 diabetes pathophysiology. *Nature* 2024;**627**:347–357.
34. Keaton JM, Kamali Z, Xie T, Vaez A, Williams A, Goleva SB, et al. Genome-wide analysis in over 1 million individuals of European ancestry yields improved polygenic risk scores for blood pressure traits. *Nat Genet* 2024;**56**:778–791.
35. Zhou W, Bi W, Zhao Z, Dey KK, Jagadeesh KA, Karczewski KJ, et al. SAIGE-GENE+ improves the efficiency and accuracy of set-based rare variant association tests. *Nat Genet* 2022;**54**:1466–1469.
36. Chen S, Francioli LC, Goodrich JK, Collins RL, Kanai M, Wang Q, et al. A genomic mutational constraint map using variation in 76,156 human genomes. *Nature* 2024;**625**:92–100.
37. Privé F, Aschard H, Carmi S, Folkersen L, Hoggart C, O'Reilly PF, et al. Portability of 245 polygenic scores when derived from the UK Biobank and applied to 9 ancestry groups from the same cohort. *Am J Hum Genet* 2022;**109**:12–23.
38. Van Gijn ME, Ceccherini I, Shinar Y, Carbo EC, Slofstra M, Arostegui JJ, et al. New workflow for classification of genetic variants' pathogenicity applied to hereditary recurrent fevers by the International Study Group for Systemic Autoinflammatory Diseases (INSAID). *J Med Genet* 2018;**55**:530–537.
39. Cosson C, Riou R, Patoli D, Niu T, Rey A, Gros Lambert M, et al. Functional diversity of NLRP3 gain-of-function mutants associated with CAPS autoinflammation. *J Exp Med* 2024;**221**:e20231200.
40. Sham PC, Purcell SM. Statistical power and significance testing in large-scale genetic studies. *Nat Rev Genet* 2014;**15**:335–346.
41. The Global Cardiovascular Risk Consortium. Global Effect of Modifiable Risk Factors on Cardiovascular Disease and Mortality. *N Engl J Med* 2023;**389**:1273–1285.
42. Braddock M, Quinn A. Targeting IL-1 in inflammatory disease: new opportunities for therapeutic intervention. *Nat Rev Drug Discov* 2004;**3**:330–340.
43. Janssen CA, Oude Voshaar MAH, Vonkeman HE, Jansen TLThA, Janssen M, Kok MR, et al. Anakinra for the treatment of acute gout flares: a randomized, double-blind, placebo-controlled, active-comparator, non-inferiority trial. *Rheumatology* 2019;**58**:1344–1352.
44. Zhou X, Fernando SM, Pan AY, Laposa R, Cullen KR, Klimes-Dougan B, et al. Characterizing the NLRP3 Inflammasome in Mood Disorders: Overview, Technical Development, and Measures of Peripheral Activation in Adolescent Patients. *Int J Mol Sci* 2021;**22**:12513.

45. Libby P. Interleukin-1 Beta as a Target for Atherosclerosis Therapy: Biological Basis of CANTOS and Beyond. *J Am Coll Cardiol* 2017;**70**:2278–2289.
46. Ridker PM, MacFadyen JG, Thuren T, Libby P. Residual inflammatory risk associated with interleukin-18 and interleukin-6 after successful interleukin-1 β inhibition with canakinumab: further rationale for the development of targeted anti-cytokine therapies for the treatment of atherothrombosis. *Eur Heart J* 2020;**41**:2153–2163.
47. Robinson G, Pineda-Torra I, Ciurtin C, Jury EC. Lipid metabolism in autoimmune rheumatic disease: implications for modern and conventional therapies. *J Clin Invest* 2022;**132**:e148552.
48. The Interleukin 1 Genetics Consortium. Cardiometabolic effects of genetic upregulation of the interleukin 1 receptor antagonist: a Mendelian randomisation analysis. *Lancet Diabetes Endocrinol* 2015;**3**:243–253.
49. Netea MG, Joosten LAB, Lewis E, Jensen DR, Voshol PJ, Kullberg BJ, et al. Deficiency of interleukin-18 in mice leads to hyperphagia, obesity and insulin resistance. *Nat Med* 2006;**12**:650–656.
50. Krishnan SM, Ling YH, Huuskens BM, Ferens DM, Saini N, Chan CT, et al. Pharmacological inhibition of the NLRP3 inflammasome reduces blood pressure, renal damage, and dysfunction in salt-sensitive hypertension. *Cardiovasc Res* 2019;**115**:776–787.
51. Amancherla K, Wells JA, Bick AG. Clonal hematopoiesis and vascular disease. *Semin Immunopathol* 2022;**44**:303–308.
52. Svensson EC, Madar A, Campbell CD, He Y, Sultan M, Healey ML, et al. TET2-Driven Clonal Hematopoiesis and Response to Canakinumab. *JAMA Cardiol* 2022;**7**:521–528.
53. Schunk SJ, Kleber ME, März W, Pang S, Zewinger S, Triem S, et al. Genetically determined NLRP3 inflammasome activation associates with systemic inflammation and cardiovascular mortality. *Eur Heart J* 2021;**42**:1742–1756.
54. Martinon F, Pétrilli V, Mayor A, Tardivel A, Tschopp J. Gout-associated uric acid crystals activate the NALP3 inflammasome. *Nature* 2006;**440**:237–241.
55. Klein AL, Imazio M, Cremer P, Brucato A, Abbate A, Fang F, et al. Phase 3 Trial of Interleukin-1 Trap Rilonacept in Recurrent Pericarditis. *N Engl J Med* 2021;**384**:31–41.
56. Senkevich K, Liu L, Alvarado CX, Leonard HL, Nalls MA, Gan-Or Z. Lack of genetic evidence for NLRP3 inflammasome involvement in Parkinson’s disease pathogenesis. *NPJ Park Dis* 2024;**10**:145.
57. Gill D, Dib M-J, Cronjé HT, Karhunen V, Woolf B, Gagnon E, et al. Common pitfalls in drug target Mendelian randomization and how to avoid them. *BMC Med* 2024;**22**:473.
58. Davey Smith G, Timpson N, Ebrahim S. Strengthening causal inference in cardiovascular epidemiology through Mendelian randomization. *Ann Med* 2008;**40**:524–541.
59. Chang CC, Chow CC, Tellier LC, Vattikuti S, Purcell SM, Lee JJ. Second-generation PLINK: rising to the challenge of larger and richer datasets. *GigaScience* 2015;**4**:7.

60. Ritchie SC, Surendran P, Karthikeyan S, Lambert SA, Bolton T, Pennells L, et al. Quality control and removal of technical variation of NMR metabolic biomarker data in ~120,000 UK Biobank participants. *Sci Data* 2023;**10**:64.
61. Astle WJ, Elding H, Jiang T, Allen D, Ruklisa D, Mann AL, et al. The Allelic Landscape of Human Blood Cell Trait Variation and Links to Common Complex Disease. *Cell* 2016;**167**:1415-1429.e19.
62. Wu T, Hu E, Xu S, Chen M, Guo P, Dai Z, et al. clusterProfiler 4.0: A universal enrichment tool for interpreting omics data. *The Innovation* 2021;**2**:100141.
63. Dolgalev I. *msigdb: MSigDB Gene Sets for Multiple Organisms in a Tidy Data Format*. 2025.
64. Benjamini Y, Hochberg Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *J R Stat Soc Ser B Methodol* 1995;**57**:289–300.
65. Mangan MSJ, Olhava EJ, Roush WR, Seidel HM, Glick GD, Latz E. Targeting the NLRP3 inflammasome in inflammatory diseases. *Nat Rev Drug Discov* 2018;**17**:588–606.